ATHENA TECH

DATA SCIENCE RESOURCES

Marketing Analytics Projects

Project 1. Email Campaign Effectiveness Analysis

Problem Statement: Marketers send thousands of promotional emails, but without proper analysis, it's hard to measure what's working and what isn't. Bounce rates, open rates, and conversions can vary drastically across campaigns.

Objective: Build a model to evaluate the effectiveness of email marketing campaigns using KPIs like open rate, click-through rate (CTR), and conversion rate.

Dataset Source: Email Campaign Dataset on Kaggle

Techniques & Concepts

- Exploratory data analysis (EDA)
- Segmentation by user behavior (opens, clicks)
- KPI calculation and visualization
- Conversion analysis using funnels and attribution

Tools & Libraries: Python, Pandas, Matplotlib, Seaborn, Tableau or Power BI (optional for dashboards)

Expected Outcome: An analytical report or dashboard showing which campaign performed best and why, with insights to improve future email strategy.

Project 2. A/B Testing Simulator for Marketing Experiments

Problem Statement: Marketers need to test variations of landing pages, headlines, or CTAs. Without statistically valid A/B testing, decisions may be based on intuition rather than data.

Objective: Simulate and evaluate A/B tests using conversion data to understand which variation performs better, using hypothesis testing and confidence intervals.

Dataset Source: Simulated web traffic or email performance datasets

Techniques & Concepts

- Hypothesis testing (Z-test, T-test)
- Confidence interval calculation
- Power and sample size analysis
- Visualization of experiment results

Tools & Libraries: Python, Scipy, Statsmodels, Pandas, Matplotlib

Expected Outcome: A simulator tool or notebook that accepts input metrics (conversions, traffic) and outputs statistically significant results for A/B testing decisions.

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3. Influencer Engagement Analytics

Problem Statement: Brands invest heavily in influencer marketing but often lack tools to measure actual impact. Engagement rate, follower quality, and ROI need to be quantified.

Objective: Analyze social media influencer performance using metrics such as likes, comments, reach, and follower growth to identify high-ROI influencers.

Dataset Source: Instagram Influencer Dataset on Kaggle

Techniques & Concepts

- Engagement rate calculation
- Follower growth trend analysis
- ROI estimation per influencer
- Cluster influencers by performance metrics

Tools & Libraries: Python, Pandas, Scikit-learn, Seaborn, Plotly, Tableau for dashboards

Expected Outcome: A ranked list or dashboard showing top-performing influencers and recommendations for future partnerships.
